Maryam Maghareh, PharmD

304-276-3132 | Maryam.maghareh97@gmail.com

EDUCATION

West Virginia University — Morgantown, WV August 2024 – Present

- **Doctor of Philosophy in Pharmaceutical and Pharmacological Sciences**

West Virginia University — Morgantown, WV August 2020 – May 2024

- **Doctor of Pharmacy *(Cum Laude)***
  - **Areas of Emphasis:**
    - Translational Research
    - College Teaching in Pharmacy

West Virginia University — Morgantown, WV August 2016 - May 2020

- **Bachelor of Science in Biology** (***Magna Cum Laude)***
  - **Minor:** Sociology

HONORS AND AWARDS

- CVS Caremark Charitable Trust Scholarship Fall 2023 – Spring 2024
- Best Poster Presentation Award Spring 2023
  - Van Liere Conference at West Virginia University
- Pharmaceutics Poster First Place Award Fall 2022
  - Regional Pharmaceutical Sciences Research Symposium
- The Charles Stump Memorial and Marvin C. Wilson Award Fall 2022 - Spring 2023
- Selected for Initiation to Research Opportunities (INTRO) Program Summer 2022
- The Michael P. Clay Scholarship Fall 2021 - Spring 2022
- Dean's List at West Virginia University Fall 2020, Spring 2021, Fall 2021, Fall 2022
  - School of Pharmacy
- International Student Scholarship Fall 2016 - Spring 2020
- President's List at West Virginia University Spring 2019, Spring 2020
- Dean's List at West Virginia University Fall 2016, Spring 2017, Fall 2018, Fall 2019

RESEARCH EXPERIENCE

West Virginia University: Huber Lab — Morgantown, WV Spring 2025 - Present

Department of Pharmaceutical Sciences

Mentor: Jason Huber, Ph.D.

- Performed neuronal cell culture experiments using HT22 cells to investigate hemin-induced cellular injury and evaluate the effect of therapeutic compounds

- Conducted in vivo studies using rodent models of intracerebral hemorrhage (ICH) and middle cerebral artery occlusion (MCAO) to investigate neurovascular injury and repair mechanisms
- Processed and prepared brain tissue for cryosectioning and histological analysis
- Carried out immunohistochemistry (IHC), Histological Staining, and Western Blot analyses to investigate molecular and cellular markers of brain injury and therapeutic response
- Utilized behavioral assays to assess motor and functional outcomes in mice

West Virginia University: Bobbala Lab — Morgantown, WV Spring 2022 - Spring 2023

Department of Pharmaceutical Sciences

Mentor: Sharan Bobbala, Ph.D.

- Developed hydratable mixed–micelle formulations
- Cultured RAW 264.7 (macrophages) and MDA-MB-BR-231 (breast cancer) cells
- Measured micelles size using dynamic light scattering (DLS) method
- Determined micelles zeta potential using electrophoretic light scattering (ELS) method
- Analyzed encapsulation efficiency of the micelles for hydrophobic drug models using spectrophotometry
- Visualized cellular uptake of the micelles using confocal microscopy and flow cytometry
- Evaluated micelles toxicity using MTT assay
- Assessed the stability of the micelles

TECHNICAL SKILLS

- Rodent handling and behavioral assessment
- Stereotaxic surgery
- Tissue collection and processing
- Immunofluorescence microscopy
- Histological staining and immunohistochemistry
- Western Blotting
- Autoclave Operation, Hamilton syringe using
- Spectrophotometry; Instrument: Spectrophotometer and absorbance, luminance, and fluorescence microplate reader | Software: SoftMax pro
- Size and zeta potential determination; Instrument: Zetasizer | Software: ZS Xplorer
- Confocal microscopy and flow cytometry
- Column Chromatography
- Microsoft Office (Word, Excel, and PowerPoint) and GraphPad (Prism)

PUBLICATIONS

- **Maghareh, M.**, Carrera, E. M., Cicala, D. S., Roknuzzaman, ASM., Sarker, R., Geldenhuys, W. J., Huber, J. D. **(2026)**. Iron-driven secondary injury after intracranial hemorrhage : mitochondrial dysfunction and ferroptosis. Front. Neurosci. 20:1838011.

- Cicala, D. S., Geldenhuys, W. J., Carerra, E. M., **Maghareh, M.,** Roknuzzaman, A. S. M., Sarker, R., Huber, J. D. **(2026)**. Mitochondria and blood-brain barrier dysfunction: a promising target for cerebrovascular disorders. Fluids and Barriers of the CNS (Accepted)

- Cicala, D. S., Habib, A. Z. S. B., Roknuzzaman, A. S. M., Carerra, E. M., **Maghareh, M.,** Sarker, R., Adjeroh, D. A., Geldenhuys, W. J., & Huber, J. D. **(2026)**. Mapping Microglial Parameters Software (MMPS): An Open-Source, User-Friendly Tool for Quantitative Microglia Morphology Analysis. *Journal of visualized experiments : JoVE*, (233), 10.3791/71566.

- Geldenhuys, W. J., Mersch, J., Cicala, D., Sarker, R., Roknuzzaman, A. S. M., **Maghareh, M.,** Carerra, E. M., Lade, D. M., Menze, M. A., Konkle, M. E., & Huber, J. D. **(2026)**. Medicinal Chemistry Review of the NEET Protein Family. *ChemMedChem*, *21*(7), e202500969.

- Russell, A., Geldenhuys, W., Huber, J., Cicala, D., **Maghareh, M.,** Dibyo, R., Sarker, R., & Miller, D. **(2026)**. Neurodegeneration in traumatic brain injury is attenuated by the MitoNEET ligand TT01001. *Proceedings of the West Virginia Academy of Science*, *98*(1).

- Hughes, K. A., Misra, B., **Maghareh, M.,** Samart, P., Nguyen, E., Hussain, S., Geldenhuys, W. J., & Bobbala, S. **(2024)**. Flash nanoprecipitation allows easy fabrication of pH-responsive acetalated dextran nanoparticles for intracellular release of payloads. *Discover nano*, *19*(1), 4.

- Hughes, K. A., Misra, B., **Maghareh, M.,** & Bobbala, S. **(2023)**. Use of stimulatory responsive soft nanoparticles for intracellular drug delivery. *Nano research*, *16*(5), 6974–6990.

POSTER PRESENTATIONS

- **Maghareh M,** Hughes K.A, Saylor S, Misra B, Bobbala S. Hydratable Powders for the Self-assembly of Mixed Micelles for Drug Delivery. Presented at the 2022 Pharmaceutical Sciences Research Symposium held at WVU Health Sciences Center and the 2023 Van Liere Conference at WVU Health Sciences Center

TEACHING EXPERIENCE

West Virginia University — Morgantown, WV Fall 2022 - Spring 2023
Department of Biology
Biology Graduate Teaching Assistant (Principles of Biology and Introductory Physiology)

- Presented pre-lab talks, simplified scientific concepts, and clarified experimental procedures
- Managed two laboratory sections and engaged with 24 students in each lab
- Held office hours every week and helped students with their questions
- Graded assigned laboratory homework

West Virginia University: — Morgantown, WV Fall 2021
School of Pharmacy
Guest Speaker (First-Year Seminar)

- Developed and presented PowerPoint slides to pre-pharmacy students and familiarized them with part of the program
- Guided the students through how to prepare for Pharmacy School Interviews

West Virginia University— Morgantown, WV Fall 2019 - Spring 2020
C. Eugene Bennett Department of Chemistry
Chemistry Teaching Assistant (Fundamentals of Chemistry)

- Presented pre-lab talks, simplified scientific concepts, and clarified experimental procedures
- Managed a laboratory section and engaged with 24 students
- Graded assigned laboratory homework
- Proctored for chemistry quizzes and exams

PHARMACY EXPERIENCE

ADVANCED PHARMACY PRACTICE EXPERIENCES (APPE)

Walgreens Pharmacy— Fairmont, WV March 2024 – April 2024
- Community Rotation

WVU School of Pharmacy — Morgantown, WV February 2024 – March 2024
- Teaching Rotation

J.W. Ruby Memorial Hospital — Morgantown, WV January 2024 – February 2024
- Acute Care - Infectious Diseases Rotation

J.W. Ruby Memorial Hospital— Morgantown, WV November 2023 – December 2023
- Acute Care - Internal Medicine Rotation

J.W. Ruby Memorial Hospital — Morgantown, WV August 2023 – September 2023
- Health System Rotation

West Virginia Drug Information Center— Morgantown, WV July 2023 – August 2023
- Systems/Populations Rotation

Mary Babb Randolph Cancer Center— Morgantown, WV June 2023 – July 2023
- Hospital Administration Rotation

WVU Heart & Vascular Institute — Morgantown, WV May 2023 – June 2023
- Ambulatory Care Rotation

INTRODUCTORY PHARMACY PRACTICE EXPERIENCES (IPPE)

J.W. Ruby Memorial Hospital — Morgantown, WV May 2022
- Health System Rotation

Kroger Pharmacy — Morgantown, WV May 2021
- Community Rotation

LEADERSHIP

American Association of Pharmaceutical Scientists — Fall 2024 – Present

- Communications Chair

Graduate Student Organization — Fall 2025 – Spring 2026

- International Student Liaison

Academy of Managed Care Pharmacy — Fall 2022 – Spring 2023

- President

American Pharmacists Association — Fall 2022 – Spring 2023

- Membership Vice President

Industry Pharmacists Organization — Fall 2022 – Spring 2023

- Recruitment Chair

VOLUNTEER EXPERIENCE

WVU COVID-19 Vaccination Clinic — Spring 2021 – Fall 2021

Student Pharmacist

- Administrated COVID-19 Vaccine to 50+ patients
- Prepared 150+ doses of Pfizer/ Moderna vaccines

WVU Medicine: **Discharge Pharmacy**— Morgantown, WV — Summer 2019

Pharmacy Student Volunteer

- Assessed the received medication for patient information: patient name, patient room number, and patient costs
- Called patient rooms to inform patients of medication readiness and processed patient payment
- Sealed prescriptions, delivered prescriptions to patient rooms, and obtained patient signatures for confirmation

WVU Medicine: Mountaineer Pharmacy — Morgantown, WV — Spring 2019

Pharmacy Student Volunteer

- Verified names and birthdates, delivered correct prescriptions, informed patients of medication readiness, and processed payments

CERTIFICATIONS

- Registered Pharmacist (RPh) License, West Virginia
- College Teaching in Pharmacy